SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

SIMATS ENGINEERING COMPUTER SCIENCE AND ENGINEERING LIST OF EXPERIMENTS

SUB CODE & NAME: CSA1712-ARTIFICIAL INTELLIGENCE

4.Write the python program for Cript-Arithmetic problem

Code:

```
"""
from itertools import permutations

letters = "SENDMORY"

for values in permutations(range(10), 8):

    S, E, N, D, M, O, R, Y = values

    if S == 0 or M == 0:
        continue

    SEND = 1000*S + 100*E + 10*N + D

    MORE = 1000*M + 100*O + 10*R + E

    MONEY = 10000*M + 1000*O + 100*N + 10*E + Y

    if SEND + MORE == MONEY:

        print("S =", S)
        print("E =", E)
        print("N =", N)
        print("D =", D)
        print("M =", M)
        print("O =", O)
        print("R =", R)
        print("Y =", Y)

        print(SEND, "+", MORE, "=", MONEY)

        break
```

Output:

```
== RESTART: C:/Users/Lekha/Appl
S = 9
E = 5
N = 6
D = 7
M = 1
O = 0
R = 8
Y = 2
9567 + 1085 = 10652
```